

Yongye(Felix) Hu

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EDUCATION

University of California, Berkeley Aug 2024 - May 2025 *Master of Engineering* | *Robotics Engineering*
Berkeley, CA

Wuhan University of Science and Technology Sep 2020 - May 2024 *Bachelor of Engineering (Honours)* |
Mechatronics Engineering Wuhan, China

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd. Sep 2023 - Mar 2024
Mechatronics Engineer Intern Wuhan, China

- Proposed and implemented a magnetically-triggered encoder-based imaging system to replace low-precision pulse triggering, achieving ± 1 mm positional accuracy and reducing point-cloud data loss by 30%.
- Developed control programs for a six-axis CRP2 robotic arm using embedded C and ladder logic, enabling automated multi-axis motion for fuel rod testing.
- Designed an STM32F103-based gateway to enable Modbus RTU communication between the PC and robotic controller, ensuring stable real-time data exchange.
- Co-developed the PC GUI using Qt and C++, integrating motion control, process visualization, and fault alerts.

Lightweight, Sustainable Remote-Mining Excavator System Jan 2021 - Jul 2023
R&D Engineer Wuhan, China

- Designed a compact ESP32-based crane system with Wi-Fi remote control, enabling low-latency communication and autonomous coordination across multiple devices, achieving over 95% task repeatability.
- Consolidated five functional boards into a single 4-layer PCB with KiCad, integrating control, power, and communication modules; reduced overall footprint by 40% and verified signal integrity via 100 MHz oscilloscope.
- Designed a multi-servo gripper system with ultrasonic radar for 360° object detection, capable of grasping up to 3 objects simultaneously.

PROFESSIONAL EXPERIENCE

L.I.F.T: A Hybrid Flying and Driving Robot Platform Sep 2024 - May 2025
Robotics Engineer & Project Status Manager Berkeley, CA

- Developed a ROS2-based robotic system integrating a simulator and MPC controller, enabling precise obstacle avoidance and full trajectory completion with real-time state logging.
- Designed and implemented a unified drivetrain supporting both aerial and ground locomotion using the same motor-gear system, reducing mechanical complexity and cost.

ROBOCON: Multi-Robot Coordination System Sep 2022 - Jul 2023
Integrated Mechanical Engineer Wuhan, China

- Led mechanical design of 3 robots for a national robotics competition with 80 teams: two autonomous systems for ring retrieval and launch, and a joystick-controlled quadruped for speed racing.
- Developed embedded C firmware for STM32F407, enabling CAN-based motor control, BLE remote communication, and PID position control with ± 2 mm tracking error for an omnidirectional mobile robot.

SKILLS

- **Programming:** C++, Python, Embedded C, Linux/UNIX, ROS2, Git, Docker, MATLAB, Qt
- **Robotics:** Motion Planning (MPC), 3D Rigid Geometry, Inverse Kinematics, PID Control, Sensor Integration, Point Cloud Processing
- **Systems:** Multi-processing (DMA/Interrupts), Real-Time Control, CAN Bus, UART, Modbus RTU, State Machines
- **Hardware:** STM32, ESP32, Arduino, Raspberry Pi, PCB Design (KiCad), PLC
- **Design:** SolidWorks, Autodesk Fusion, Inventor, Ansys (FEA/Fluent), 3D Printing, CNC